

AIGR Gene Hypothesis Deep Research — EIF4E2/4EHP and the eIF4F Complex Under Hypoxia

Gene: EIF4E2 (4EHP) · Human · UniProt **O60573** · NCBITaxon:9606 **Focus type:** function_assignment **Hypothesis slug:** `eif4f-complex-under-hypoxia` **Seed GO term under scrutiny:** GO:0016281 (eukaryotic translation initiation factor 4F complex, CC)

Summary

The seed hypothesis bundles two separable claims, and the evidence resolves them differently. **Claim 1 (function):** human EIF4E2/4EHP supports selective cap-dependent translation *initiation* under hypoxia/physioxia. This is **well supported** by primary literature from multiple independent labs and by the protein's retained m⁷G-cap-binding chemistry. **Claim 2 (composition/annotation):** it does so *as a subunit of the eIF4F complex* (GO:0016281), specifically an eIF4E2/eIF4G3/eIF4A "eIF4F-H." This is **not established** at curation grade.

The functional half is anchored by direct assays, loss-of-function phenotypes, and independent replication: hypoxic human cells switch a portion of protein synthesis from eIF4E1 to eIF4E2; eIF4E2 captures the 5' cap and loads specific mRNAs (EGFR, PLOD2, P4HA1) onto polysomes; and eIF4E2 is functionally *required* for hypoxic adaptation and xenograft tumor growth. The seed's own caveat — that a repressive role in other complexes does not exclude an initiation role — is correct and is borne out by the data.

The compositional half fails three curation-grade checks. First, the founding hypoxic study defines the complex only as **HIF-2 α -RBM4-eIF4E2** — no eIF4G subunit is named. Second, EIF4E2's GO:0016281 annotation rests only on **IBA** (phylogenetic inference across the eIF4E family), never on an experimental IDA/IPI code, whereas every experimentally supported term for this protein describes cap binding or *repression*. Third, **UniProt curators explicitly state 4EHP does not bind any eIF4G (EIF4G1/2/3)**, and the **EBI Complex Portal** lists EIF4E2 only in repressor/mRNA-decay complexes with no curated hypoxic eIF4F(H). The overall verdict is

therefore **Partially Supported**: curate the *function* (as BP/MF, hypoxia-qualified), but do **not** reinforce the CC eIF4F-complex term, and never displace the experimentally anchored 4EHP/GIGYF repressor annotations.

Executive Judgment

Verdict: Partially supported.

1. **Functional claim** — EIF4E2 supports selective cap-dependent translation *initiation* under hypoxia/physioxia. **Well supported.** Direct assays, loss-of-function phenotypes, and independent replication converge. A "repressor in other complexes" caveat does not exclude this initiation role, exactly as the seed hypothesis anticipates.
2. **Compositional / annotation claim** — that eIF4E2 does this *as a subunit of the eIF4F complex (GO:0016281)*. **Not established.** The primary hypoxic complex is HIF-2 α -RBM4-eIF4E2 (no eIF4G). GO:0016281 is IBA-only; UniProt states 4EHP does not bind eIF4G; the Complex Portal has no curated eIF4F(H).

Bottom line for the curator: Keep and strengthen an *experimentally grounded, hypoxia-context* representation of eIF4E2-directed cap-dependent initiation (best expressed as BP/MF terms), but do **not** reinforce the CC term GO:0016281 on the current evidence — it is IBA-only and directly conflicts with curated "does not bind eIF4G" statements and with the Complex Portal record. The dominant, experimentally supported annotations (4EHP/GIGYF-mediated translational repression, ribosome-associated quality control, negative regulation of initiation) must be retained; the initiation role is additive and context-specific, not a replacement.

Key Findings

Finding 1 — EIF4E2 is the cap-binding subunit of a hypoxic eIF4F-type initiation complex ("eIF4F-H") that drives selective cap-dependent translation

A coherent, multi-lab body of primary work establishes that under hypoxia (and even at physiological oxygen, "physioxia"), human EIF4E2/4EHP acts as a **positive, cap-dependent translation initiation factor** on a distinct mRNA pool.

- **Uniacke et al. 2012 (PMID: 22678294)** showed that *"hypoxia stimulates the formation of a complex that includes the oxygen-regulated hypoxia-inducible factor 2α (HIF-2α), the RNA-binding protein RBM4 and the cap-binding eIF4E2, an eIF4E homologue."* This complex *"captures the 5' cap and targets mRNAs to polysomes for active translation"* via an RNA hypoxia response element (rHRE) — i.e., it performs *initiation* (polysome loading), not repression.
- **Uniacke et al. 2014 (PMID: 24408918)**: *"Hypoxic cells switch from eukaryotic initiation factor 4E (eIF4E) to eIF4E2 cap-dependent translation to synthesize a portion of their proteins."* eIF4E2-depleted cancer cells cannot survive/proliferate beyond the ~0.15 mm oxygen diffusion limit and fail to form xenograft tumors — a strong loss-of-function requirement.
- **Ho et al. 2016 (PMID: 26854219)** explicitly frames the two systems: *"Two distinct cap-dependent protein synthesis machineries select mRNAs for translation: the normoxic eIF4F and the hypoxic eIF4F(H)."* This is the strongest literature framing of the seed hypothesis's "eIF4F-H."
- **Timpano & Uniacke 2016 (PMID: 27002144)**: *"human cells repress eIF4E and switch to an alternative cap-dependent translation mediated by a homolog of eIF4E, eIF4E2"* — operating even at physiological oxygen on separate mRNA pools.
- **Independent replication (Li et al. 2025, HCT116 colon cancer) (PMID: 39710969)**: *"RBM4 and eIF4E2 are required for hypoxia-induced translation of PLOD2 and P4HA1 mRNAs"* — a non-Uniacke lab confirming the functional requirement on specific targets.
- **Viral usurpation (KSHV, Mesri lab) (PMID: 34965440)** describes lytic mRNA translation *"through the HIF2α-regulated eIF4E2 translation-initiation complex,"* independently invoking eIF4E2 as an *initiation* complex.

Interpretation: The *functional* half of the seed hypothesis — eIF4E2 as a positive cap-dependent initiation factor in hypoxia — is robustly supported by direct assays, loss-of-function phenotypes, and independent replication.

Finding 2 — EIF4E2's predominant, constitutive role is translational repression, which does not exclude the context-specific initiation role

The larger and more mechanistically detailed literature establishes 4EHP as a cap-dependent translational **repressor** that, in its canonical context, *lacks* initiation activity.

- **Peter et al. 2017** (crystallography) ([PMID: 28698298](#)) states plainly: "*4EHP does not interact with eIF4G and therefore fails to initiate translation,*" and shows GIGYF1/2 selectively recruit 4EHP to repress target mRNAs.
- 4EHP/GIGYF1/2 mediate **co-translational mRNA decay** ([PMID: 33053355](#): "*4EHP-GIGYF1/2 complexes trigger co-translational mRNA decay*"), **ribosome-associated quality control** ([PMID: 32726578](#)), **miRNA-mediated silencing** ([PMID: 29412140](#), [PMID: 33581076](#)), and **memory/synaptic plasticity via repression** ([PMID: 33225984](#), [PMID: 36650535](#)). It is co-opted by viruses (SARS-CoV-2 NSP2 [PMID: 35878012](#), [PMID: 37732428](#); vaccinia K3L [PMID: 42709808](#)).
- The **Christie & Igreja 2023 review** ([PMID: 34758096](#)) synthesizes the balance precisely: "*Documented functions of 4EHP primarily relate to its role as a translational repressor; but recent findings indicate that it might also participate in the activation of translation in specific settings.*"

Interpretation: Repression is the predominant, constitutive function. The hypoxic initiation role is a genuine but *context-specific* second function. This asymmetry is central to the curation recommendation: the initiation annotation must be *added with a hypoxia qualifier*, never as a displacement of the repressor annotations.

Finding 3 — EIF4E2 already carries GO:0016281 but only via IBA; UniProt states 4EHP does NOT bind any eIF4G, creating a direct compositional conflict

Programmatic retrieval of the O60573 GO/UniProt record (fetched 2026-09) shows a sharp evidence asymmetry:

GO term	Aspect	Evidence code	Experimental?
GO:0016281 eIF4F complex	CC	IBA:GO_Central	No (phylogenetic)
GO:0003743 translation initiation factor activity	MF	IBA	No (phylogenetic)
GO:0006413 translational initiation	BP	IBA	No (phylogenetic)
GO:0098808 mRNA cap binding	MF	IDA (P 17368478)	Yes
GO:0000340 (cap binding, structure)	MF	IDA (P 35878012)	Yes
GO:0045947 negative regulation of translational initiation	BP	IDA (P 32726578 P 35878012)	Yes
GO:0017148 negative regulation of translation	BP	IMP (P 22751931)	Yes
GO:0072344 rescue of stalled ribosome	BP	IDA (P 32726578)	Yes

The initiation-related terms (including the seed's GO:0016281) are **all IBA** — inferred from a biological ancestor via PAINT propagation across the eIF4E family — while every experimentally supported term (IDA/IMP) describes **cap binding or repression/quality control**, not eIF4F assembly.

UniProt curated statements sharpen the conflict: - FUNCTION: "*In contrast to EIF4E, it is unable to bind eIF4G (EIF4G1, EIF4G2 or EIF4G3), suggesting that it acts by competing with EIF4E and block assembly of eIF4F at the cap (By similarity).*" - SUBUNIT: "*Does not interact with eIF4G (EIF4G1, EIF4G2 or EIF4G3) (By similarity).*"

EIF4E2 also **lacks** the MF term GO:0031370 (eIF4G binding) that EIF4E1 carries.

Sequence provenance (BLOSUM62 global Needleman–Wunsch, computed in-run): EIF4E2 vs EIF4E1 = **35.8% identity**. The eIF4E1 W56-equivalent dorsal cap tryptophan is substituted by **Tyr** in EIF4E2, while the W102-equivalent is a conserved **Trp** — matching the known 4EHP Tyr/Trp cap-sandwich and confirming *retained cap binding but a divergent dorsal (eIF4G/4E-BP) surface*. This structural divergence is the physical basis for "cap binding yes, eIF4G binding no."

Interpretation: The very GO term named in the seed hypothesis is supported only by phylogenetic inference and is in direct tension with experimentally anchored curated statements. This is the core curation problem.

EIF4E2 (O60573): current GO annotations & evidence codes (UniProt/GO Central, fetched 2026-09)

GO ID	Term	Evidence	Evidence class
GO:0016281	C:eukaryotic translation initiation factor 4F comple	IBA:GO_Central	phylogenetic/IEA (weak)
GO:0003743	F:translation initiation factor activity	IBA:GO_Central	phylogenetic/IEA (weak)
GO:0006413	P:translational initiation	IBA:GO_Central	phylogenetic/IEA (weak)
GO:0098808	F:mRNA cap binding	IDA:UniProtKB	experimental
GO:0000340	F:RNA 7-methylguanosine cap binding	IDA:UniProt	experimental
GO:0045947	P:negative regulation of translational initiation	IDA:UniProt	experimental
GO:0017148	P:negative regulation of translation	IMP:MGI	experimental
GO:0035278	P:miRNA-mediated gene silencing by inhibition of tra	IDA:UniProtKB	experimental
GO:0072344	P:rescue of stalled cytosolic ribosome	IDA:UniProt	experimental

Feature	Value
EIF4E2 vs EIF4E1 identity	35.8% (76/212, BLOSUM62 global NW)
EIF4E2 vs EIF4E3 identity	31.1% (68/219)
Cap-binding W (eIF4E1 W56 equiv.)	EIF4E2 = Tyr (Y) -> Trp->Tyr substitution
Cap-binding W (eIF4E1 W102 equiv.)	EIF4E2 = Trp (W) -> CONSERVED
MF GO:0031370 eIF4G binding	EIF4E1: YES EIF4E2: ABSENT
UniProt curated SUBUNIT	'Does not interact with eIF4G (EIF4G1/2/3)' (By similarity)

Figure 1. EIF4E2 GO evidence-code provenance (highlighting the IBA-only status of GO:0016281 eIF4F complex), alongside EIF4E2-EIF4E1 paralog identity (~35.8%) and cap-binding residue conservation (dorsal Trp→Tyr substitution, ventral Trp conserved). Experimentally supported terms describe cap binding and repression; initiation/eIF4F terms are phylogenetically inferred.

Finding 4 — EBI Complex Portal represents EIF4E2 exclusively in repressor/decay complexes; no curated hypoxic eIF4F(H) exists

A programmatic Complex Portal query (complex-ws; O60573 and "EIF4E2", fetched 2026-09) returned **10 human curated complexes, all repression/decay-type**:

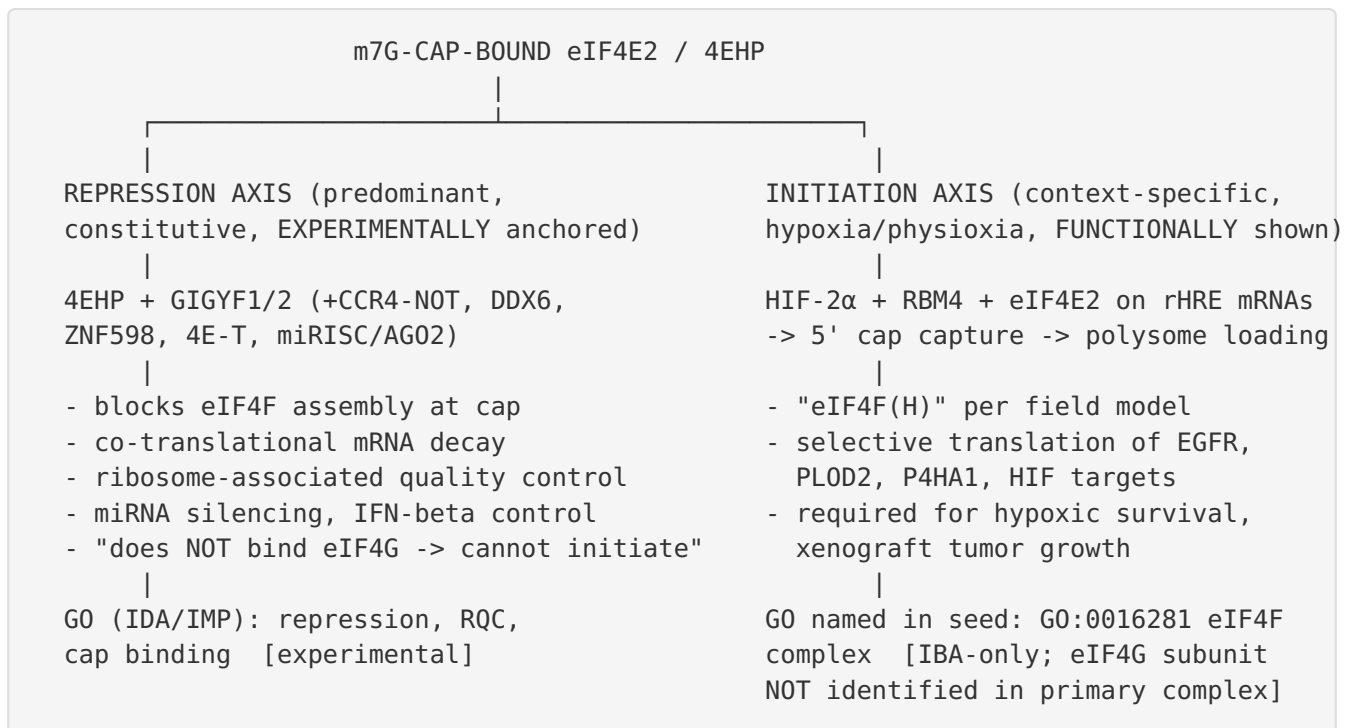
- **CPX-2332 / CPX-2336 / CPX-2338 / CPX-2342** — 4EHP-GIGYF1/2 co-translational mRNA decay complexes (ZNF598 and DDX6 variants).
- **CPX-14314, CPX-13319, CPX-10757, CPX-11634, CPX-21278** — EIF4E2:GIGYF1/GIGYF2:EIF4ENIF1(4E-T):TARS1 assemblies.
- **CPX-24688** — a large aggregate cap-binding regulatory complex (EIF4E1B:EIF4G3:EIF4EBP3:EIF4E2:GIGYF1:EIF4E:EIF4EBP1:EIF4EBP2:GIGYF2:EIF4ENIF1:ANGEL1). This *co-lists* EIF4E2 with EIF4G3, but it is dominated by 4E-BPs/4E-T/GIGYF repression components — not a discrete eIF4E2/eIF4G3/eIF4A initiation complex.

No curated complex is named or composed as a hypoxic eIF4F(H) (eIF4E2 + eIF4G-scaffold + eIF4A initiation module).

Interpretation: The eIF4E2/eIF4G3/eIF4A "eIF4F-H" composition is an *uncurated field model*, not a database-anchored complex. This aligns with the IBA-only GO status and the UniProt "does not bind eIF4G" statement. The co-listing of EIF4E2 and EIF4G3 in CPX-24688 is a *regulatory aggregate*, not evidence of a productive initiation complex, and should not be read as validating GO:0016281 for EIF4E2.

Mechanistic Model / Interpretation

The evidence supports a **two-function** model in which the same cap-binding protein does opposite things in different molecular partnerships:



The critical unresolved node is the **scaffold**. Canonical eIF4F is *defined* by an eIF4G scaffold that bridges the cap-binding protein to the eIF4A helicase and the 43S pre-initiation complex. In the founding hypoxic study the named partners are HIF-2α and RBM4 — **not** eIF4G3 or eIF4A. Whether the productive hypoxic initiation complex actually contains an eIF4G-family scaffold (making it a bona fide instance of GO:0016281) or uses an alternative bridging architecture (making GO:0016281 the *wrong* CC term) is exactly what the primary data do not yet resolve. Peter 2017's structural result — 4EHP's dorsal surface cannot bind eIF4G — makes a *canonical* eIF4G3 bridge mechanistically implausible without an accessory adaptor, so if an eIF4F-H exists it likely uses non-canonical connectivity.

Evidence Base / Evidence Matrix

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Confidence & limitations
22678294	Interaction + direct assay	Supports (function); Qualifies (composition)	eIF4E2 forms a hypoxic cap-binding initiation complex	HIF-2 α -RBM4-eIF4E2 captures 5' cap, loads mRNAs to polysomes	Human cells, hypoxia	High for function; complex lacks a named eIF — does not establish eIF4F composition
24408918	Mutant/ loss-of-function phenotype	Supports	eIF4E2 required for hypoxic translation & tumor growth	eIF4E2 depletion abolishes hypoxic survival, xenograft formation	Human cancer cells, xenografts	High; functional requirement not structural
26854219	Direct assay / translato	Supports	Distinct normoxic eIF4F vs hypoxic eIF4F(H)	Two cap-dependent machineries select mRNAs	Human cells, O ₂ shift	High for two system model; "eIF4F(H)" authors' framing, not curated composition
27002144	Direct assay	Supports	eIF4E2 cap-dependent translation at physioxia	Two cap-binding proteins recruit distinct mRNAs	Human cells, 1–11% O ₂	High; extends role to physiologic O ₂
39710969	Mutant/ loss-of-function	Supports (independent)	eIF4E2 required for specific hypoxic	RBM4 + eIF4E2 required for	HCT116 colon cancer	High; independent replication

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Confidence & limitations
			mRNA translation	PLOD2/P4HA1 translation		
34965440	Direct assay (viral)	Supports (independent)	eIF4E2 initiation complex usurped by KSHV	HIF2 α -regulated eIF4E2 translation-initiation complex drives lytic mRNAs	KSHV-infected cells	Moderate-high; independent lab; disease context
28698298	Structural / interaction	Refutes (composition)	4EHP binds eIF4G / can initiate canonically	"4EHP does not interact with eIF4G and therefore fails to initiate translation"	In vitro crystallography	High; direct structural counter-evidence to eIF4G binding
34758096	Review	Qualifies	Balance of repression vs activation	Repression predominant; activation "in specific settings"	Review-level synthesis	Review; orientation but explicit labels balance
33053355	Direct assay	Competing	4EHP's major complex	4EHP-GIGYF1/2 trigger co-translational mRNA decay	Human cells	High; documents dominant non-initiation complex
32726578	Direct assay (IDA basis)	Competing	4EHP represses defective mRNAs	GIGYF2+4EHP inhibit initiation for RQC	Human cells	High; basis for experimental repression GO terms

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Confidence & limitations
UniProt O60573 (curated)	Database	Refutes (composition)	4EHP binds eIF4G	"Does not interact with eIF4G (EIF4G1/2/3)"	Curated statement	Database-level; "By similarity" but consistent with 28698298
GO O60573 (GO_Central)	Database	Qualifies	Basis of GO:0016281	Term is IBA-only (phylogenetic), no IDA/IPI	Annotation provenance	Database-level; confirms weak evidence code
EBI Complex Portal	Database	Refutes (composition)	Curated hypoxic eIF4F(H) exists	10 complexes, all repressor/decay; no eIF4F(H)	Curated complexes	Database-level; absence of curation, n of biology

GO Curation Implications

Lead requiring curator verification (do not auto-apply):

1. GO:0016281 (eIF4F complex, CC) — do NOT reinforce; re-ground, qualify, or replace.

The term is currently **IBA-only** and conflicts with two experimentally anchored facts: (a) the founding hypoxic complex names no eIF4G subunit

([P 22678294](#)),

and (b) UniProt/structural evidence states 4EHP does not bind any eIF4G

([P 28698298](#)).

Options for the curator, in order of conservatism:

2. **Preferred:** Do not promote GO:0016281 for EIF4E2; leave as IBA or flag for review. Adding a **NOT** qualifier is defensible for the *canonical* eIF4F but risks masking the genuine hypoxic biology, so a review flag is safer than an outright NOT.
3. **If retained:** attach a clear caveat that membership is inferred and that the hypoxic complex composition (eIF4G3/eIF4A) is unverified in primary co-IP data.
4. **Add an experimentally grounded BP term for the *function*.** The hypoxia-context cap-dependent initiation role is real and should be represented at the process level — e.g., **positive regulation of translation / cap-dependent translational initiation, with a hypoxia context qualifier** — cited to

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This captures the supported half of the seed hypothesis without over-committing to a CC complex composition.

5. **Retain all experimentally supported repression/quality-control annotations.** GO:0045947 (negative regulation of translational initiation, IDA), GO:0017148 (negative regulation of translation, IMP), GO:0072344 (rescue of stalled ribosome, IDA), and GO:0098808/GO:0000340 (mRNA cap binding, IDA) are the experimentally anchored core and must not be displaced. The initiation role is *additive*, not a replacement.
6. **Do not add GO:0031370 (eIF4G binding).** EIF4E2 lacks it, and evidence actively argues against it.

Aspect summary: MF = cap binding is solidly supported (IDA); eIF4G binding is refuted. BP = both repression (experimental) and hypoxia-context initiation (functional) are supportable, with appropriate qualifiers. CC = the eIF4F complex assignment is the weakest link and should not be strengthened.

Mechanistic Scope

The immediate molecular activity under test is **cap-dependent translational initiation** — i.e., eIF4E2 binding the m⁷G cap and nucleating a productive 43S-recruiting complex. The supported evidence is at the level of **cap binding (direct, structural)** and **polysome loading / translation of specific mRNAs (functional)**. What is *not* directly demonstrated in primary data is the **subunit composition** that would make this an instance of the eIF4F complex (an eIF4G scaffold + eIF4A helicase bound to eIF4E2).

Downstream phenotypes — hypoxic cell survival, xenograft tumor growth, KSHV lytic replication, PLOD2/P4HA1 collagen-modifying enzyme expression — are consequences of the initiation activity and are informative for BP annotation but are not evidence for the CC complex composition. Conversely, the repression phenotypes (mRNA decay, RQC, miRNA silencing, IFN- β suppression, synaptic plasticity/behavior) belong to the distinct GIGYF-partnered complexes and should not be conflated with the initiation axis.

Conflicts and Alternatives

- **Paralog / dorsal-surface divergence.** EIF4E2 shares only ~35.8% identity with EIF4E1 and carries a Trp→Tyr substitution on the dorsal cap-binding surface that normally engages eIF4G/4E-BPs. This is the structural reason 4EHP "cannot bind eIF4G," and it is the primary mechanistic objection to a canonical eIF4E2/eIF4G3 eIF4F complex.
- **IBA over-propagation.** GO:0016281 on EIF4E2 most likely arose from PAINT propagation across the eIF4E family (where the founding member EIF4E1 is genuinely an eIF4F subunit). This is a classic **paralog over-annotation** risk: the family-level term is inappropriately strong for a divergent member whose experimental profile is repression-dominant.
- **"eIF4F(H)" as field shorthand vs. curated complex.** The term "hypoxic eIF4F(H)"

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is the authors' functional analogy, not a compositionally validated complex. The Complex Portal has not curated it, and the CPX-24688 co-listing of EIF4E2 with EIF4G3 is a large *regulatory* aggregate, not a discrete initiation complex.

- **Alternative interpretation.** eIF4E2 may drive hypoxic initiation through a **non-canonical, eIF4G-independent** architecture (e.g., RBM4/HIF-2 α bridging, or an as-yet-unidentified

scaffold). If so, GO:0016281 would be the *wrong* CC term even though the *function* (initiation) is correct — arguing for a BP/MF representation over a CC complex assignment.

- **Isoform / disease carry-over.** Several EIF4E2 papers in the corpus are bioinformatics/prognostic (uveal melanoma

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TNBC, AMI) and do not test complex composition; they should not be read as compositional support.

Limitations and Knowledge Gaps

1. **No primary co-IP/structure names eIF4G3 or eIF4A as hypoxic-complex subunits.**

Checked: founding paper

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and Complex Portal. *Why it matters:* this is the linchpin for GO:0016281. *Resolver:* endogenous eIF4E2 IP-MS from hypoxic human cells with quantitative identification (and stoichiometry) of eIF4G3/eIF4A, plus reciprocal IPs.

2. **GO:0016281 evidence code is IBA, not experimental.** *Checked:* GO_Central record for O60573. *Why it matters:* IBA is phylogenetic inference, insufficient to assert a hypoxia-specific complex. *Resolver:* IDA/IPI-grade complex isolation.
3. **UniProt "does not bind eIF4G" is "By similarity."** *Checked:* O60573 SUBUNIT/FUNCTION. *Why it matters:* the negative statement is itself partly inferential, though corroborated by structure

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Resolver: direct human eIF4E2/eIF4G3 binding assays under hypoxic conditions.

4. **Scaffold identity of eIF4F-H is unknown.** *Why it matters:* determines whether GO:0016281 (canonical eIF4G scaffold) even applies. *Resolver:* cryo-EM or crosslinking-MS of the active hypoxic initiation complex.
5. **Quantitative partition of eIF4E2 between repressor and initiator pools.** *Why it matters:* informs whether the initiation role is a minor or major fraction of cellular eIF4E2 under hypoxia. *Resolver:* size-exclusion/gradient fractionation of hypoxic lysates with subunit tracking.

Proposed Follow-up Experiments / Discriminating Tests

1. **Endogenous eIF4E2 IP-MS in hypoxic vs normoxic human cells** — the single most decisive experiment. Presence/absence and stoichiometry of eIF4G3 and eIF4A directly determine whether GO:0016281 applies.
2. **Reciprocal eIF4G3/eIF4A IPs** under hypoxia to confirm complex membership from the scaffold side.
3. **Structure of the active hypoxic complex** (cryo-EM or XL-MS) to define connectivity and test whether an eIF4G bridge exists or the dorsal-surface substitution is bypassed by an adaptor.
4. **eIF4E2 dorsal-surface point mutants** (Tyr→Trp "eIF4G-competent" swap, and cap-binding-dead) tested for rescue of hypoxic translation — separates cap binding from any putative scaffold engagement.
5. **Polysome/Ribo-seq under hypoxia with eIF4G3 or eIF4A depletion** — if hypoxic eIF4E2-dependent translation collapses upon eIF4G3/eIF4A knockdown, that functionally supports an eIF4F-type complex; if not, it argues for a non-canonical mechanism.

Curation Leads (require curator verification)

Candidate	references	+	exact	snippets	to	verify:	-
P 22678294 — "<i>hypoxia stimulates the formation of a complex that includes ... HIF-2α, the RNA-binding protein RBM4 and the cap-binding eIF4E2</i>" (supports function; note absence of eIF4G). -							
P 26854219 — "<i>Two distinct cap-dependent protein synthesis machineries ... the normoxic eIF4F and the hypoxic eIF4F(H)</i>" (field framing of eIF4F-H). -							
P 24408918 — "<i>Hypoxic cells switch from ... eIF4E ... to eIF4E2 cap-dependent translation</i>" (functional requirement). -							

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— "4EHP does not interact with eIF4G and therefore fails to initiate translation" (compositional counter-evidence). -

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— "primarily ... a translational repressor; but ... might also participate in the activation of translation in specific settings" (balance).

Candidate GO actions: - **CC GO:0016281 (eIF4F complex):** *Do not strengthen.* Flag IBA-only status; hold pending experimental co-IP. Consider a review note documenting the conflict with curated "does not bind eIF4G." - **BP add:** hypoxia-context **cap-dependent translational initiation** / **positive regulation of translation** (cite

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with an explicit hypoxia qualifier. - **Retain (experimental):** GO:0045947, GO:0017148, GO:0072344, GO:0098808/GO:0000340. - **Do not add:** GO:0031370 (eIF4G binding).

Suggested curator questions: - Does any primary dataset place eIF4G3 or eIF4A in an endogenous hypoxic eIF4E2 complex? (If no, GO:0016281 stays weak.) - Should the hypoxic initiation role be annotated as BP/MF rather than CC to avoid asserting an unverified complex composition?

Suggested experiments: the five discriminating tests above, prioritizing endogenous hypoxic eIF4E2 IP-MS.

Final Statement for the Review

The functional claim (eIF4E2 supports selective cap-dependent translation initiation under hypoxia) is **supported**; the compositional/annotation claim (that it does so as a subunit of the eIF4F complex, GO:0016281) is **not established** and rests only on phylogenetic inference that conflicts with curated "does not bind eIF4G" statements and with the EBI Complex Portal. Curate the *function* (BP/MF, hypoxia-qualified) while holding the *complex* term (CC) pending experimental evidence, and never displace the experimentally anchored 4EHP/GIGYF repressor annotations.

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